Determine Distance on the Coordinate Plane

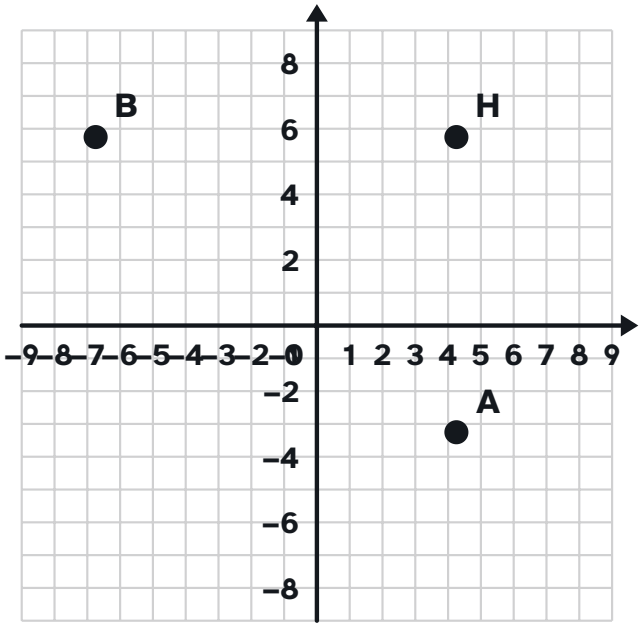
Find two distances on the coordinate plane and compare them.

Name _____ **Date** _____

Tonight's spark

A park map has a hub at G, (1.75, 8.25). Cruz walks straight to C, (1.75, 2.25). Dara walks straight to D, (–5.25, 8.25). One unit is one block. Who walks farther?

Guess first: Cruz goes down and Dara goes left. Who walks farther?



Our class model: find the shared coordinate, then use the other

Try it

Work it out on paper. Write one answer on each line.

1 **Which coordinate do (3, 1) and (3, –5) share?**

A. The x-coordinate **B.** The y-coordinate **C.** Neither one

My answer _____

2 **Find the distance between (2, 6) and (2, 1).**

A. 7 **B.** 5 **C.** 6 **D.** 1

My answer _____

3 **Find the distance between (–4, 3) and (5, 3).**

A. 1 **B.** 4 **C.** 5 **D.** 9

My answer _____

4 **Find the distance between (1.5, 4) and (1.5, –2.5).**

A. 6.5 **B.** 1.5 **C.** 4 **D.** 2.5

My answer _____

5 **Ana walks from (0, 5) to (0, –3). Beto walks from (0, 5) to (6, 5). Who walks farther?**

A. Beto, by 4 units **B.** Ana, by 2 units **C.** Ana, by 14 units **D.** They walk the same

My answer _____

Talk about it

Which coordinate does each trip share? Do you add or subtract for each trip, and who goes farther?

Home mission

Cross two forks on the table to make the axes. Put one coin 3 finger-widths above the crossing and another 5 below it. Ask your student: how far apart are the coins? Add or subtract? Why?

If we get stuck, ask

- Which coordinate do these two points share?
- Do the other two values have the same sign or opposite signs?
- Does the trip cross an axis? Then what do we do?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/7-6-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it